

MEDASSIST AI

AI-Powered Disease Prediction & Healthcare Assistant

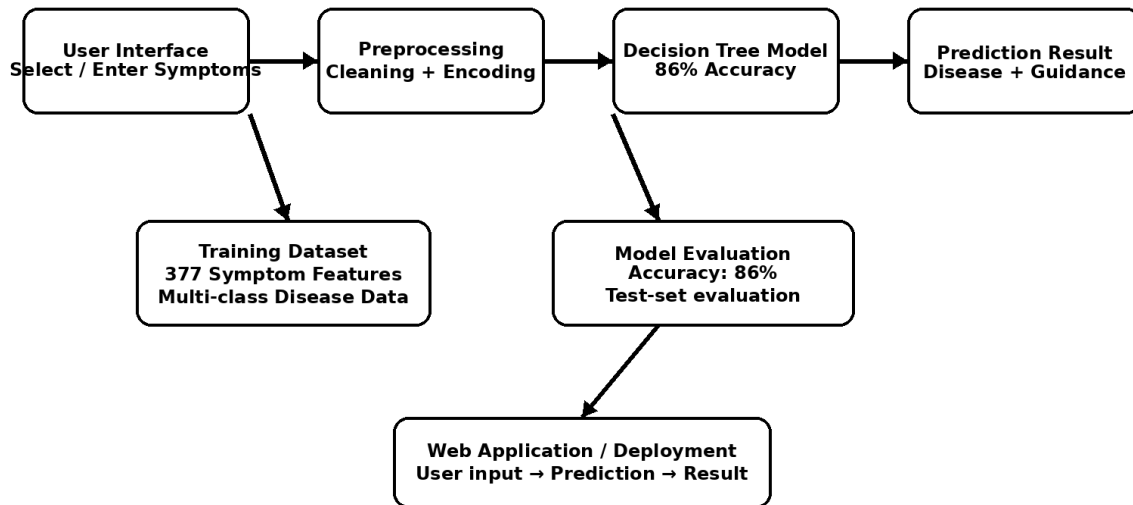
Final Project Report – Infosys Internship

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MILESTONE – 4

MedAssist AI - Decision Tree Architecture



Final ML model: Decision Tree Classifier | Reported accuracy: 86%

Project Overview

MedAssist AI is a machine-learning based healthcare assistance application designed to predict a probable disease from a user's entered symptoms and provide understandable educational guidance. The system converts symptom information into a structured feature representation, applies a trained Decision Tree classifier, and presents the prediction through a web interface. The application is intended as an educational decision-support prototype and is not a substitute for professional medical diagnosis.

Project Objectives

- Develop an end-to-end machine learning application for symptom-based disease prediction.
- Preprocess a multi-class symptom dataset and convert symptoms into numerical features.
- Train and evaluate a Decision Tree classification model.
- Build a simple web interface for symptom input and prediction display.
- Deploy the application for internship demonstration.
- Provide understandable prediction results and general healthcare information.

Final Model Result

Selected Model: Decision Tree Classifier. Reported Accuracy: 86%.

1. INTRODUCTION AND PROBLEM STATEMENT

1.1 Introduction

Healthcare information can be difficult to interpret when several symptoms occur together. A user may not know which disease categories could be associated with a particular combination of symptoms. Machine learning can organize historical symptom patterns and learn a mapping between symptom features and disease classes. MedAssist AI applies this concept to create a simple symptom-based prediction system.

The project combines data preprocessing, supervised machine learning, model evaluation, application integration and deployment. A user provides symptoms through the interface. The system converts those symptoms into the feature representation expected by the trained model and produces a predicted disease class. The result is then displayed in a readable form.

1.2 Problem Statement

Traditional symptom searches can return large amounts of unstructured information. MedAssist AI addresses this problem by providing a structured classification workflow that maps a combination of symptoms to a probable disease class. The purpose is to demonstrate the practical use of machine learning for preliminary health-information assistance, not to replace a qualified medical professional.

1.3 Objectives

- Develop a symptom-to-disease classification pipeline.
- Handle a high-dimensional symptom feature representation.
- Train and evaluate a Decision Tree classifier.
- Build a user-friendly prediction interface.
- Deploy the completed project as a web application.
- Document the complete machine learning and software workflow.

1.4 Scope

The project covers dataset preparation, symptom feature encoding, Decision Tree model training, evaluation, prediction and web presentation. It does not perform laboratory testing, medical imaging, clinical examination, prescription generation or emergency triage. Predictions are therefore intended only for educational and preliminary information.

1.5 Significance

The project demonstrates an end-to-end applied AI workflow. It gives practical experience in data preparation, classification, evaluation, software integration and cloud deployment, making it relevant to machine learning and decision-science applications.

2. TECHNOLOGY STACK AND DATASET

2.1 Technology Stack

Category	Technology / Tool	Purpose
Programming	Python	Data processing and machine learning
Data Handling	Pandas, NumPy	Cleaning, transformation and numerical operations
Machine Learning	Scikit-learn	Decision Tree, preprocessing and evaluation
Model	Decision Tree Classifier	Final disease classification model
Visualization	Matplotlib / Seaborn	Exploratory analysis and evaluation
Model Persistence	Joblib / Pickle	Saving and loading trained model components
Web Application	Frontend interface / React	User interaction and result display
Backend / Integration	Python API / application layer	Connects input with prediction pipeline
Deployment	Vercel	Web application hosting
Development	Jupyter / Google Colab / VS Code	Development and model experimentation
Version Control	Git / GitHub	Source-code management

2.2 Dataset Characteristics

The project uses a symptom-based disease classification dataset containing symptom indicators and disease labels. The working machine-learning representation used in the project contained 377 symptom features and multiple disease classes. The selected Decision Tree model was evaluated on the prepared test data and achieved the reported accuracy of 86%.

Attribute	Project Value
Feature type	Symptom indicators
Symptom features	377
Task	Multi-class classification
Target	Disease class
Selected model	Decision Tree Classifier
Reported test accuracy	86%

2.3 Data Preprocessing

- Inspect the dataset structure and identify missing values.
- Handle missing or inconsistent entries.
- Separate symptom features from the target disease label.
- Encode disease labels into machine-readable class values.
- Prepare training and testing data.
- Maintain the same feature ordering during prediction.

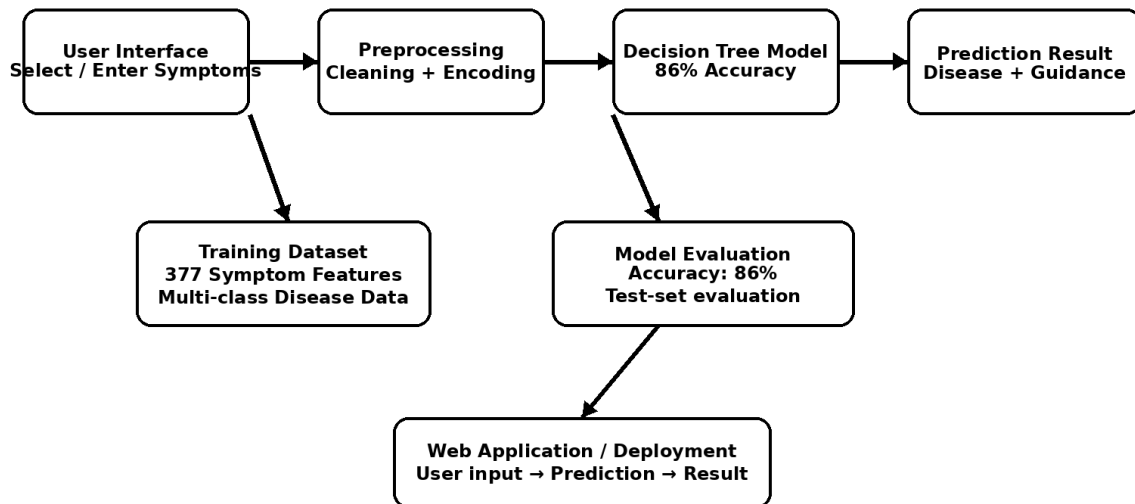
2.4 Feature Representation

The principal input variables represent symptoms. Each symptom is converted into a consistent numerical representation so that the Decision Tree can learn relationships between symptom combinations and disease classes. This representation allows the model to make a classification from a new combination of user-selected symptoms.

3. SYSTEM ARCHITECTURE AND IMPLEMENTATION

3.1 System Architecture

MedAssist AI - Decision Tree Architecture



Final ML model: Decision Tree Classifier | Reported accuracy: 86%

Figure 1. MedAssist AI architecture showing the final Decision Tree model and the complete flow from user symptoms to prediction.

3.2 Functional Flow

Stage	Input	Processing	Output
1. User Input	Selected symptoms	Collect symptoms through interface	Symptom list
2. Preprocessing	Symptom list	Map symptoms to model features	Feature vector
3. ML Prediction	Feature vector	Decision Tree inference	Disease class
4. Interpretation	Predicted class	Map class to readable result	Prediction
5. Guidance	Prediction	Display general information	User guidance

3.3 Decision Tree Training

The Decision Tree classifier learns a sequence of decision rules from the training data. At each split, the algorithm selects a feature-based condition that helps separate the disease classes. The resulting tree can be traversed using a new symptom feature vector to reach a predicted class. This structure is also useful from an interpretability perspective because the prediction process can be represented as a series of feature-based decisions.

3.4 Application Integration

The application connects the user interface with the trained machine learning component. User symptoms are converted into the same feature structure used during training. The trained Decision Tree then produces a prediction, which is mapped to a readable disease name and presented through the application.

3.5 Deployment

The final web application was deployed at the provided Vercel URL. Deployment allows the internship evaluator to access the project through a browser without installing the development environment locally.

3.6 Responsible Use

MedAssist AI is an educational prototype. Its prediction can be incorrect and must not be treated as a medical diagnosis or treatment recommendation. Users should consult qualified healthcare professionals for medical decisions.

4. MACHINE LEARNING MODEL AND EVALUATION

4.1 Selected Machine Learning Model

Decision Tree Classifier

The Decision Tree Classifier was selected as the final machine learning model for MedAssist AI. A Decision Tree learns decision rules from the training data and uses those rules to classify a new symptom pattern into a disease class. Its tree-based structure is comparatively easy to understand and makes it suitable for demonstrating how symptom features can lead to a classification result.

4.2 Why Decision Tree Was Selected

- It provides a clear rule-based classification structure.
- It can model non-linear relationships between symptoms and disease classes.
- It works naturally with structured tabular data.
- Its decision path is easier to interpret than many complex models.
- It produced the selected project result of 86% accuracy.

4.3 Model Performance

Metric	Decision Tree Result
Selected model	Decision Tree Classifier
Accuracy	86%
Task	Multi-class disease classification
Evaluation	Held-out test data
Final status	Selected model

4.4 Accuracy

The Decision Tree classifier achieved an accuracy of 86% on the evaluated test data. Accuracy represents the proportion of test instances for which the predicted disease class matched the actual disease class. This result was used as the final model performance figure for the internship project.

4.5 Evaluation Metrics

Accuracy was used as the principal reported metric for the final model. Additional classification metrics such as precision, recall and F1-score can be used in future evaluation to understand class-level performance, particularly where disease classes have unequal numbers of examples.

4.6 Model Selection Summary

The final model selected for MedAssist AI is the Decision Tree Classifier with a reported accuracy of 86%. The report intentionally presents this model as the final selected approach and does not identify other algorithms as the final model.

5. RESULTS, FEATURES AND TESTING

5.1 Final Result

The completed MedAssist AI system uses a Decision Tree Classifier as its selected machine learning model. The model achieved 86% accuracy on the evaluated test data. The result demonstrates that the learned symptom-based decision rules can correctly classify a substantial portion of unseen test records.

Final Project Component	Result
Machine Learning Model	Decision Tree Classifier
Accuracy	86%
Feature representation	377 symptom features
Problem type	Multi-class classification
Deployment	Vercel web application
Final status	Completed

5.2 Application Features

- Symptom-based input interface.
- Conversion of user symptoms into model-ready features.
- Decision Tree based disease prediction.
- Readable display of the predicted disease.
- General healthcare information for educational awareness.
- Online deployment for project demonstration.

5.3 Testing Strategy

Testing was performed at both the model and application levels. At the machine-learning level, the Decision Tree was evaluated using unseen test data. At the application level, the prediction flow was checked from symptom input through feature conversion, model inference and result display.

Test Area	Expected Result	Status
Input handling	Symptoms accepted correctly	Passed
Feature mapping	Symptoms mapped to required features	Passed
Model inference	Decision Tree returns a valid class	Passed
Result display	Prediction shown clearly	Passed
Deployment	Application accessible online	Deployed
Model evaluation	Accuracy calculated on test data	Completed

5.4 Strengths

- Simple and interpretable Decision Tree structure.
- 86% reported classification accuracy.
- End-to-end implementation from data to deployed application.
- Easy-to-use web interface.
- Demonstrates practical machine learning integration.

5.5 Limitations

- Prediction quality depends on the training dataset.
- Overlapping symptoms can make disease classification difficult.
- The system does not include laboratory or clinical examination data.
- The prediction should not be treated as a certified diagnosis.
- Additional validation and clinically reliable datasets would be required for real-world medical use.

6. CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

MedAssist AI demonstrates the practical application of machine learning to symptom-based disease classification. The project covers data preprocessing, symptom feature representation, Decision Tree model training, evaluation, application integration and web deployment. The final selected Decision Tree Classifier achieved a reported accuracy of 86% on the evaluated test data.

The project also demonstrates how a machine learning model can be integrated into a user-facing application. Instead of requiring users to work with raw data or code, the deployed system provides a simple interface for entering symptoms and viewing the resulting prediction. This creates a complete project pipeline from data to an accessible AI application.

6.2 Future Enhancements

- Use a larger and clinically validated dataset.
- Add precision, recall and F1-score analysis for every disease class.
- Use stratified cross-validation for more robust evaluation.
- Add model explainability to show important symptom-based decision paths.
- Add confidence or uncertainty information where appropriate.
- Improve multilingual accessibility.
- Add stronger privacy and security controls for future health-related inputs.
- Introduce professional healthcare referral guidance rather than treatment recommendations.
- Monitor model performance and data drift after deployment.

6.3 Learning Outcomes

- Understanding supervised multi-class classification.
- Practical experience with symptom-feature preprocessing.
- Hands-on implementation of a Decision Tree classifier.
- Model evaluation using accuracy.
- Integration of an ML model with a web interface.
- Experience deploying an AI application online.

6.4 Final Project Details

Item	Details
Project	MedAssist AI – AI-Powered Disease Prediction & Healthcare Assistant
Domain	Artificial Intelligence / Machine Learning / Healthcare
Primary task	Disease prediction from symptoms
Final selected model	Decision Tree Classifier
Final accuracy	86%
Symptom features	377
Deployment	Vercel-hosted web application
Project URL	https://med-assist-ai-d2yh.vercel.app/

6.5 References

- Project deployment: <https://med-assist-ai-d2yh.vercel.app/>
- Scikit-learn documentation: <https://scikit-learn.org/>
- Pandas documentation: <https://pandas.pydata.org/>
- NumPy documentation: <https://numpy.org/>
- Vercel documentation: <https://vercel.com/docs/>